

ALBION, NE, USA

WMO: 723441

Lat: 41.730N

Lon: 98.054W

Elev: 548

StdP: 94.91

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54921

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.0	-17.2	-24.0	0.5	-18.8	-21.6	0.6	-15.8	15.0	-3.4	13.6	-3.0	2.8	330	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	33.9	23.0	32.3	22.6	30.8	22.2	25.3	30.6	24.3	29.7	23.4	28.8	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.7	19.8	29.0	22.6	18.5	28.0	21.9	17.7	27.1	80.7	30.8	76.4	29.9	72.7	29.0	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.7	9.1	DB	-25.2	36.6	2.8	1.6	-27.2	37.8	-28.8	38.7	-30.4	39.6	-32.4	40.7	(4)
(5)			WB	-25.4	26.9	2.7	1.0	-27.4	27.6	-29.0	28.2	-30.6	28.8	-32.5	29.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-4.0	-2.9	4.2	9.6	15.6	21.5	23.8	22.1	18.1	10.4	3.7	-3.0	(6)
(7)		DBStd	11.17	6.71	6.75	6.43	5.36	4.98	3.51	3.20	2.96	4.38	5.03	5.57	6.21	(7)
(8)		HDD10.0	1730	433	361	198	70	9	0	0	0	2	56	198	402	(8)
(9)		HDD18.3	3528	692	594	437	264	116	11	3	5	57	251	439	661	(9)
(10)		CDD10.0	1730	0	1	20	58	181	345	429	374	244	68	9	0	(10)
(11)		CDD18.3	487	0	0	1	3	29	106	173	121	50	4	0	0	(11)
(12)		CDH23.3	4947	0	1	18	58	367	1090	1771	1040	527	72	4	0	(12)
(13)		CDH26.7	1828	0	0	3	14	124	392	748	361	166	20	0	0	(13)
(14)	Wind	WSAvg	3.8	3.9	4.0	4.5	4.8	4.3	3.7	2.8	2.7	3.3	3.7	3.7	3.7	(14)
(15)	Precipitation	PrecAvg	657	11	18	46	63	102	101	84	83	65	41	31	13	(15)
(16)		PrecMax	954	39	76	181	243	222	244	242	285	194	106	109	30	(16)
(17)		PrecMin	369	0	0	0	12	10	22	10	11	11	0	0	1	(17)
(18)		PrecStd	140	10	15	41	45	49	59	53	52	43	31	27	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.9	18.8	26.0	28.6	33.0	35.0	36.5	34.8	32.9	29.9	23.0	14.9	(19)
(20)			MCWB	7.6	9.1	13.2	16.5	19.0	22.4	23.2	23.4	21.5	18.6	12.0	7.2	(20)
(21)		2%	DB	10.5	13.5	21.8	24.8	30.0	32.7	34.5	32.3	30.9	25.0	19.1	11.1	(21)
(22)			MCWB	4.9	7.2	12.0	14.2	19.0	21.9	23.4	23.6	21.6	15.4	10.7	5.9	(22)
(23)		5%	DB	7.3	10.0	18.0	22.0	27.4	31.0	32.6	30.2	28.6	22.4	15.9	7.8	(23)
(24)			MCWB	3.2	5.1	11.0	13.3	17.3	21.3	23.6	23.1	20.5	14.1	9.0	3.9	(24)
(25)		10%	DB	4.8	6.3	14.8	19.2	24.9	28.9	30.9	28.7	26.6	19.9	12.6	5.1	(25)
(26)			MCWB	2.0	2.8	9.0	11.9	16.5	20.6	23.3	22.1	19.0	12.9	7.4	2.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	10.1	16.4	17.6	21.7	25.7	26.9	26.4	24.2	20.1	14.0	8.9	(27)
(28)			MCDWB	14.7	17.0	21.1	24.2	28.9	31.4	32.5	31.4	29.6	25.6	19.9	12.9	(28)
(29)		2%	WB	5.2	7.3	14.0	15.7	20.3	23.9	25.6	25.0	22.8	17.4	11.6	6.0	(29)
(30)			MCDWB	9.9	13.4	19.4	22.4	27.3	29.9	31.1	30.1	28.2	22.7	18.1	10.6	(30)
(31)		5%	WB	3.5	5.1	11.4	14.1	18.9	22.8	24.7	23.9	21.7	15.4	9.7	4.1	(31)
(32)			MCDWB	6.7	9.2	17.3	20.0	25.2	28.7	30.2	28.8	26.5	19.7	15.5	7.5	(32)
(33)		10%	WB	2.1	3.1	9.0	12.7	17.4	21.7	23.7	23.0	20.5	13.7	7.7	2.4	(33)
(34)			MCDWB	4.7	6.1	14.9	18.2	23.5	27.0	29.3	27.4	25.0	18.6	12.4	5.0	(34)
(35)	Mean Daily Temperature Range	MDBR	10.8	11.1	12.5	12.7	12.4	12.1	12.0	11.5	13.5	13.5	13.0	10.5	(35)	
(36)		5% DB	MCDBR	14.6	16.2	18.7	17.4	16.2	14.2	14.0	13.5	15.1	18.2	18.5	14.6	(36)
(37)			MCWBR	9.9	10.1	10.2	9.1	7.2	6.0	6.1	6.7	7.1	9.4	10.6	10.0	(37)
(38)		5% WB	MCDWB	13.5	15.5	16.3	14.8	13.7	12.3	12.0	11.8	12.5	13.9	17.0	13.7	(38)
(39)			MCWBR	9.6	10.1	9.6	8.4	6.7	6.3	6.2	6.5	6.7	8.3	10.3	9.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.274	0.282	0.307	0.355	0.376	0.395	0.400	0.404	0.363	0.324	0.281	0.270	(40)	
(41)		taud	2.462	2.475	2.443	2.320	2.346	2.339	2.345	2.321	2.393	2.484	2.516	2.478	(41)	
(42)		Ebn at Noon	889	936	945	914	899	879	870	856	871	872	874	863	(42)	
(43)		Edh at Noon	74	86	101	123	123	124	122	122	105	85	70	67	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.09	3.00	3.96	5.04	5.88	6.71	6.70	5.72	4.64	3.28	2.31	1.75	(44)	
(45)		RadStd	0.18	0.17	0.34	0.28	0.41	0.45	0.31	0.39	0.30	0.35	0.17	0.17	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page